

Unit-2

Software Requirement Process

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- Requirement Engineering Process
- Feasibility Studies
- Requirements Elicitation and Analysis
- Requirements Validation
- Requirement Management

Requirement Engineering Process

Four high-level activities

- Feasibility study - Assessing the system usefulness to the business
- Elicitation and Analysis - Discovering requirements
- Specification - Converting the requirements into standard form
- Validation - Checking that the requirements actually define the system that the customer wants.

Spiral RE process

- Activities are organized around a spiral with the output being a system requirement document.
- Time and effort devoted to each activity in each iteration depends on the stage of the overall process
- Early stage in spiral, more effort required to understand FR, NFR, UR and SR.
- Later stage in spiral, more effort devoted to elicitation and SR.

Fig : Spiral view of RE process

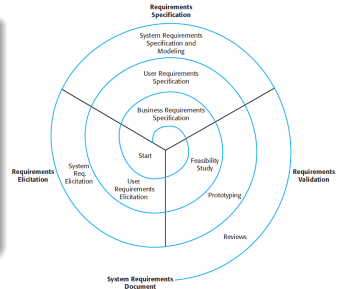
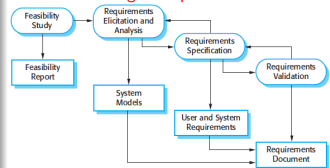


Fig : RE process



Case study-1

- Apply the spiral RE process and determine the time, effort for the activities in case study.
- Consider the average productivity rate 50FP/PM with time in hours.

Feasibility Studies

- Can we build software to meet the defined scope? Is the project feasible ?
- The main aim of the feasibility study activity is to determine whether it would be financially and technically feasible to develop the product.
- This study involves the analysis of the problem and collection of all relevant information relating to the product.
- Different data items collected - input to the system, the processing, output data, constraints on the behaviour of the system.
- A feasibility report enables the organization to decide whether the project can be taken up by the organization or not.

This study should answer the following questions: sample provided

- Can current software and hardware technologies can be used for development ?
- Is the proposed system will be cost-effective from a business point of view ?
- Can we develop within existing budgetary constraints ?

Types of feasibility

- Technical feasibility : can the system be implemented within schedule and budget using current technology?
- Operational feasibility : does the system contribute to the overall objectives of the organization?
- Economic feasibility : can the system be integrated with other systems that are used?

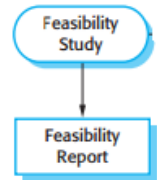


Fig : Feasibility Study

Feasibility report(Contd..)

Case study-2

- Consider Android software code, identify function in a context and determine the any three technical, operational and economic feasibility.

Case study-3

- Download a software code from Github, prepare any FIVE technical questionnaire pointing out challenges and solution for it.

Requirement Elicitation and analysis

What is Elicitation ?

- Elicitation is an activity in RE process to find out the services, constraints, performance of the proposed system.
- It has iterative 4 process and ends with SRS.

What is 4 process of Elicitation ?

- Requirement discovery, requirement classification & organization, requirement prioritization & negotiation, requirement specification

Difficulties in Elicitation

- Unrealistic demand, Requirement Engineering(RE) experience item Discovery commonalities and conflict requirement, Political factors, New requirements from new stakeholders

Stakeholders in Elicitation

- System Stakeholders, Development Engineer, Maintenance Engineer, Business Manager, Domain experts, Trade union representatives

Fig : Elicitation and Analysis in RE process

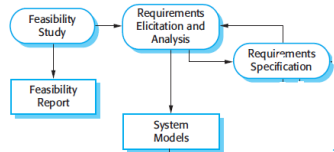
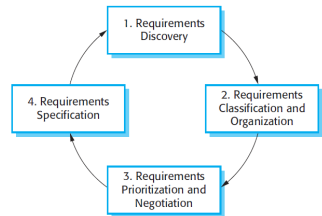


Fig :The Requirements Elicitation and Analysis Process



Requirement Elicitation and analysis (contd..)

Requirement discovery

• Basic guidelines

- 1 Meeting with stakeholders
- 2 Draft an agenda for idea collection
- 3 A facilitator appointed to control the meeting
- 4 A mechanism (eg. chat room, virtual, forum, wall stickers, work sheets) to store the requirement is defined

• Req discovery Goals

- 1 Identify the problem
- 2 Propose elements of the solution
- 3 Negotiate different approach
- 4 Specify a preliminary set of solution

Requirement discovery

• Facilitator goals

- 1 Invite stakeholders by fixing the schedule
- 2 Collect the product request from stakeholders
- 3 Collect the propose elements using mechanism
- 4 Check for ambiguity, omissions, and errors in requirements

Req classification & organization

• Basic functions

- 1 Translate the requirements into technical requirements for the proposed software
- 2 Identifies three requirements :
Normal, expected, exciting

Req classification & organization

• Types of requirements

- 1 Normal requirement - req that satisfies the customer eg. graphic display
- 2 Expected requirement - req that creates dissatisfaction eg. reliability
- 3 Exciting requirement - req that go beyond the customer expectations eg. prevent vulnerability attack

• Methods used

- 1 Open interview, closed interview
- 2 Observation, Survey, historical data

• Mechanism used

- 1 Diagrams, Metrics

Requirement Elicitation and analysis (contd..)

Requirement Classification & Organization

● Usage scenario

- 1 Translate the technical requirements into technical SE activities
- 2 Scenarios are created using use cases.
- 3 Use case diagram is developed.

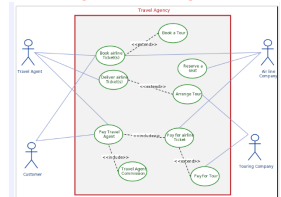
Requirement Prioritization & Negotiation

- Balancing the business benefit of requirement against its cost and product
- concentrates on user and system requirements
- focus on effort
- trade-off among conflicting goals
- allocate resources

Req specification - SRS

- Use SRS IEEE template to document the requirements.
- Use natural language to write specification
- NF and NFR should be stated.
- User requirements and system requirements should be clear, unambiguous, easy to understand, complete, consistent.
- User requirements can be supported by diagrams and tables.
- System requirements can include graphical system model or mathematical system model.
- Not to include architecture diagrams or design diagram
- A specific architecture diagram for complex large system should be included.

Fig :Use case diagram



Use case diagram

- Case study - Booking the tour through touring company.
- Use case diagram represents the proposed software (Ref:Fig).

Requirement Elicitation Techniques

- ① Interviewing - open, closed
- ② Scenarios
- ③ Use cases
- ④ Ethnography

Case study-4

- Consider Android software code, identify few functions in a context and do the following.
 - **Discovery** : Identify the problem, propose a solution, negotiate different approach and specify preliminary set of solution
 - **Classification** : State the requirements in terms of normal, expected and exciting. Draw the use case diagram.
 - **Priorization and Negotiation** : Find the conflicting requirements and provide trade-off solution.
 - **Specification** : A specific diagram with the various functionality to be addressed in the software.

Case Study-5

- Video editing system: The system accepts digital video as input, stores the video on disk, and then allows the user to do a widerange of edits to the digitized video. The result can then be output to DVD or other media.

Video editing system:

- Draw the use case diagram.

Case Study-6

- Define the user and system requirements and draw the use case diagram for any one of the following activities:
 1. Making a withdrawal at an ATM.
 2. Using your charge card for a meal at a restaurant.
 3. Buying a stock using an on-line brokerage account.
 4. Searching for books (on a specific topic) using an on-line bookstore.

Requirement Elicitation and analysis (contd..)

Case Study-7: Web-based pothole tracking and repair system (PHTRS)

- Citizens can log onto a website and report the location and severity of potholes. As potholes are reported they are logged within a public works department repair system and are assigned an identifying number, stored by street address, size (on a scale of 1 to 10), location (middle, curb, etc.), district (determined from street address), and repair priority (determined from the size of the pothole). Work order data are associated with each pothole and include pothole location and size, repair crew identifying number, number of people on crew, equipment assigned, hours applied to repair, hole status (work in progress, repaired, temporary repair, not repaired), amount of filler material used, and cost of repair (computed from hours applied, number of people, material and equipment used). Finally, a damage file is created to hold information about reported damage due to the pothole and includes citizens name, address, phone number, type of damage, and dollar amount of damage. PHTRS is an online system; all queries are to be made interactively.
- Draw the Use case diagram for PHTRS.

What is req validation?

- Checks the defined system requirements.
- It discover errors in the request to prevent rework cost after system development.
- Provides solid foundation for design.
- Inconsistencies, omissions, and ambiguity are examined.

Sample Checklist

- Is each requirement bounded and unambiguous ?
- Is each requirement testable once implemented?
- Do any requirement conflict with other requirement ?

Types of Checks

- Validity check - Whether diverse stakeholders requirements are satisfied?
- Consistency checks - No contradictory constraints in the requirements document.
- Completeness checks - FR and NFR should be stated in SRS.
- Realism checks - Can the requirements be implemented within the budget and schedule.
- Verifiability - Is the requirements are testable ?

Validation techniques

- Requirement reviews - check for errors and inconsistencies.
- Prototyping - an executable model is experimented to validate the requirements.
- Test case generation - reveals the requirement problem by testing the requirements.

Case study-8

- Consider Android software code, identify a few functions in a context and do the following.
 - **Validity** Give a diverse requirement and state how it is satisfied.
 - **Consistency** Give a contradictory constraints in the requirement and state how it is satisfied.

Requirement Management

What is req management?

- Process of understanding and controlling changes to system requirement and to track the requirement to maintain.

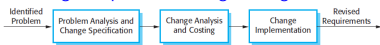
What is req traceability?

- Keep track of the relationship between requirements, their sources, and the system design, so as analyze the proposed changes and their impact in other parts of the system.

Types of changing requirement

- **Enduring requirement** : Requirement that are associated with the core, slow to change activities of an organization.
- **Volatile requirement** : Requirements are likely to change due to supporting activities of the organization.

Fig : Requirement Change Management



Why requirements change?

- Change is possible for large system.
- Requirement are bound to be incomplete.
- Requirement must evolve for changes.

Why requirements change is inevitable ?

- Business and technical environment change
- Adding new requirement for user support to meet the proposed system goals.
- Discover requirements to support diverse user community.

Requirement management process:

- **Requirement management planning** : Decide on requirement identification, change management process, traceability policies, tool support (requirement storage).
- **Requirement change management** : Manages the new requirement and is responsible to create a revised requirement document. It involves 3 stages as given in Fig.

Case study-9

- Consider Android software code, identify few functions in a context and do the following.
 - **Traceability** Track a requirement and its impact in other parts of the software. Prepare a track table.
 - **Requirement change** Specify a requirement change and its impact in the software.

- Activities of RE process
- Feasibility report
- Elicitation and analysis process.
- Draw use case diagram.
- Validation techniques
- Requirement management process

- Prepare a report on technical, operational and economic feasibility for the proposed moodle software.
- Draw the use case diagram for the proposed moodle software.
- Identify the normal, expected, and exciting requirements for the proposed moodle software.
- Apply the elicitation process to collect the requirement for moodle software. **Note: apply 4 steps:discovery, classification, proritization & negotiation, specification**
- Apply the any one elicitation techniques to collect the requirement for moodle software.

- Pressman, Roger S. Software Engineering: A Practitioner's Approach. New York: McGraw-Hill, 2010. 7th Edition
- Sommerville, Ian., Software Engineering 9th edition, Addison-Wesley